Aarush Agarwal

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GITHUB https://github.com/kratos999-athena

Professional Summary

3rd-year Computer Science B.Tech student with a strong foundation in Generative AI (GraphRAG, LangChain), machine learning, and advanced algorithms. Award-winning developer experienced in architecting end-to-end AI systems, including medical text summarizers, multi-LLM support frameworks, and highly rigorous data pipelines. Seeking to apply analytical problem-solving and AI expertise to drive innovation in advanced research and core software development.

Education & Credentials

Bharati Vidyapeeth College of Engineering **2023-2027**
Computer Science Engineering, CGPA: 9.27
Sardar Patel Vidyalaya **2023**
CBSE Class XII, 94.2%
Sardar Patel Vidyalaya **2021**
CBSE Class X, 93.2%

Professional Experience

VARAHE ANALYTICS
DATA SCIENCE INTERN **June 2025- Aug 2025**

- Engineered highly rigorous data processing pipelines utilizing Python Pandas to execute granular, booth-level electoral mapping, enforcing strict data validation protocols to achieve 100% accuracy with zero margin for error in voter distribution analytics.
- Architected and deployed an automated data aggregation system to synthesize complex political metrics into comprehensive candidate dossiers, eliminating manual bottlenecks and successfully scaling the rapid production of hundreds of high-fidelity profiles within hours.
- Designed and maintained dynamic business intelligence dashboards via Google Looker Studio, transforming massive, complex voter datasets into actionable visual insights enabling automated generation of 150+ candidate analytical profiles within 2 hours

Technical Skills

- **Programming Languages:** Python, SQL, C++, Java, JavaScript, C, Cypher
- **AI & Machine Learning:** LangChain, LangGraph, RAG, GraphRAG, Scikit-learn, TensorFlow, Hugging Face, Computer Vision & NLP
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Google Looker Studio, Advanced Excel
- **Databases & Vector Stores:** PostgreSQL (Relational), Neo4j (Graph), ChromaDB (Vector)
- **Developer Tools & Cloud:** Git/GitHub, Docker, Streamlit, Google Cloud (Service Accounts), FastAPI, HTML, CSS
- **Core Engineering Concepts:** Data Structures & Algorithms, Multithreading, OOP, System Design, Client-Server Architecture

Projects

GraphRAG Medical Text Summarizer | Tech: *Python, Cypher, LangChain, GraphRAG, Neo4j, ChromaDB, Prompt-Engineering*
GitHub: https://github.com/kratos999-athena/MedGraph-COT

- Architected a robust medical text summarization engine utilizing GraphRAG and LangChain to process and synthesize highly complex, unstructured clinical data improving clinical completeness by 94.1% across benchmarked documents
- Improved entity-level recall from 0.18 to 0.34 and achieved 97.3% coverage win rate over baseline summarization systems and achieved 66% clinical deployability rate

Multi-LLM Agent System for Alzheimer's Support | Tech: *Python, LangChain, LangGraph, OpenCV, HTML, CSS, JavaScript, GraphRAG*
GitHub: https://github.com/kratos999-athena/Alzheimer-Patient-Help

- Designed an interactive, multi-agent support framework utilizing LangGraph to assist Alzheimer's patients with cognitive retention and daily social interactions.
- Integrated real-time facial recognition with conversational LLM agents capable of identifying individuals enabling context-aware personalized responses and adaptive memory reinforcement during live interactions.

Scan2Sheet: Advanced OCR Data Pipeline | Tech: *Python, OpenCV, PaddleOCR, Pandas, Docker*
GitHub: https://github.com/kratos999-athena/Scan2Sheet

- Developed custom parsing algorithms to dynamically detect and reconstruct complex table geometries across 100+ heterogeneous PDF, minimizing redundant OCR calls through intelligent region detection and layout-aware preprocessing.
- It autonomously extracts unstructured data from PDFs and converts it into fully formatted Excel spreadsheets reducing external OCR/API usage by 60% through selective parsing and batching strategies

Achievements & Extracurriculars

- **Awards**: 1st Place at Innovate and Automate Hackathon by ISA Delhi (Hardware) & Techshuttle Debugging Competition; 2nd Place in Pinnacle (Academic Excellence)
- **Leadership**: General Secretary of Blissful Minds Society (BMS), directing organizational strategy, managing technical operations, and leading student initiatives.